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School Timetabling by Computer a Technical History

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School Timetabling by Computer

a Technical History

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Summary
The purpose of this report is to give a brief description of the development of computer-based timetabling systems. We begin with an outline of the nature of the timetabling problem, suggesting the kind of requirements that arise in practice and the criteria used to evaluate timetables. The next three sections outline the various systems that have been developed, the early systems which were largely unsuccessful, the second generation systems which have been successfully implemented, and the latest systems which try new approaches to the problem, not always successfully.

Introduction
In this paper we give a brief description of the development of computer-based school timetabling systems in the UK. Our purpose is not to provide detailed evaluations of each system, still less to suggest a preference for any particular system, but rather to outline the complex nature of the problem involved and the different approaches which people have adopted to tackle the problem.

The school timetable is one of the most important aspects of a school's administration because it reflects the extent to which a headmaster can satisfy a large number of complex educational requirements within the constraints imposed by available resources – manpower, rooms, equipment, etc. However, for medium-size or larger schools, finding a timetable which best achieves the objectives is a difficult problem which can involve six to eight weeks of arduous work by a senior member of staff, so it seems natural to ask whether computers can take over the timetabling task.

To see the role that computers might play in timetabling it is worth considering the timetabling process.

The timetabling process
It is usual to divide the timetabling process into two stages. The first is the curriculum planning stage, in which the headmaster sets out the broad requirements he would like his timetable to satisfy. This is usually the result of detailed discussions with heads of departments to determine factors such as organization of option groups, number of periods of each subject to be taught to each class, allocation of staff and rooms to particular classes, etc. This is the most difficult stage of timetabling, partly because the planning process involves complex balancing of a large number of educational considerations, and partly because the planning has to be conducted with considerable uncertainty about pupil option choices and staff availability. At this stage there is little detailed checking of whether the requirements can be met with the resources the headmaster expects to have available, beyond a check that no class or department will be required to teach or be taught for more than the number of periods available.

The second stage occurs when a firm set of requirements has been agreed, and the school tries to see whether these can be fitted together to make a workable timetable. This stage represents the detailed checking of the feasibility of the requirements specified in stage one, and it is a tedious and time-consuming activity when done by hand.

It is generally agreed that the role of computers is to handle the second stage of timetabling, the detailed construction of a timetable from a headmaster's basic requirements. Viewed this way, computer-aided timetabling has some fairly obvious advantages.

1) It will save a senior member of staff several weeks of tedious effort, freeing him for more important tasks.
2) It may encourage more frequent timetabling; a survey of Scottish schools (Lawrie, 1969) indicated that ten per cent of schools left their timetables unaltered for long periods of time. In a situation of rapidly changing ideas in education, this could be undesirable. Schools may even timetable more than once a year, allowing them to adapt to changing staffing availabilities or to have more outdoor activities in the summer term, etc.
3) By providing a detailed check on the feasibility of the headmaster's requirements, it will allow the headmaster to experiment with different forms of curriculum organization to see which he likes best. Thus the computer can help, indirectly, in the first stage of timetabling.

This represents the conventional view of computer-assisted timetabling, a view which underlines the approaches of the earlier timetabling systems. The next section will describe the kind of requirements the headmaster is expected to supply for such systems and the criteria used to assess the resulting timetables. Before discussing these points, however, it is worth noting one major disadvantage of using computers in the timetabling process. In practice the complexity of the timetabling problem makes it impossible for the headmaster to specify completely the requirements he wants for a timetable, and he will rely on his hand timetabler to make appropriate decisions as the timetable itself is constructed. As an outsider to the school, the computer system cannot make these choices, and so will usually require the headmaster to make a more detailed list of requirements than he would like. This suggests that there is a closer link between the two timetabling stages than was suggested above, and hence timetabling systems should allow the school to interact closely with the computer in timetable construction. It is this view of the timetabling process that underlies the latest computer-based timetabling systems, which we shall discuss later.

The timetabling problem

In this section we discuss the kind of requirements which it is reasonable to expect that a headmaster can specify and a computer can handle. These are as follows:

(i) *Simple Requirements* A simple requirement states that a particular teacher and a particular class must meet together for a particular number of periods in a week. These simple requirements are the most basic form of requirements in a timetable problem. We shall assume that we have a *requirements matrix R*, whose i-j entry indicates that number of periods a week for which class i meets teacher j. It can be shown that a timetabling problem in which all requirements are of this form has a solution which is easily found. It is the existence of other requirements, which act as further constraints on R, which make timetabling non-trivial. These other requirements include:

(ii) *Preassignments.* These state that a specific teacher/class meeting must occur on a certain period of the week. A special case of this is a non-availability requirement – e.g. a teacher does not come in on Monday morning.

(iii) *Preference Preassignments.* This states that a certain teacher/class meeting must occur within a certain section of the week, e.g. Wednesday afternoon, without stating specific periods within the section.

(iv) *Multiple Periods.* Certain teacher/class meetings must occur on a number of consecutive periods (which may or may not be allowed to span certain breaks).

(v) *Setting requirements.* Certain teacher/class meetings must occur simultaneously; this allows the register classes to be broken down into different units according to interests or abilities. Option groups are a special case of this. In some schools the entire upper school will be scheduled in this way.

(vi) *Special Rooms.* Certain teacher/class requirements must occur in a special room, e.g. a physics laboratory. Lunch rooms are a special case of this.

Requirements (i) – (vi) can be thought of as constraints on individual teacher/class meetings, and timetables are evaluated by the percentage of such requirements that are met; this is referred to as the *percentage* fill.

The other requirements concern the *appearance* of the timetable as a whole.

(vii) *Balance of subjects over week.* If a particular teacher meets a particular class for, say, four periods a week, these should be evenly spaced over the week, unless otherwise stated.

(viii) *Balance between subjects.* Certain subjects are similar in certain respects and should not occur together – e.g. PE and Sport.

(ix) *Miscellaneous.* These include requirements such as that a particular teacher (class) should not have to walk far between classes. A special case of this is posed by split-site schools.

Requirements (vii) – (ix) are more aesthetic, and rough criteria have been evolved to assess the *quality* of timetables.

These are the requirements which it is usual to specify for a timetabling problem.

In the next section we describe the first attempts to tackle the problem set out above. These are described in some detail, since they form the basis for other approaches.

The first generation systems

All these approaches take the conventional view of the timetabling problem, and so require the headmaster to prepare a complete list of requirements of the kind outlined in the previous section. From this list a timetable is generated. These systems were not successful operationally, at least in their original formulation, but they laid the foundations on which later systems were based.

A. The Teddington system (Appleby *et al.* 1960) This approach is based on two heuristic rules that one would expect a hand timetabler to utilize – (i) start scheduling those requirements for which you have least scope for manoeuvre; (ii) where there is some degree of choice, assign a requirement to a period in which it is likely to interfere minimally with other requirements of the timetable. The program handles double periods, balance over the week requirements, special room requirements and setting requirements. The input to the program is a number of *lines* stating that a given class and teacher must meet for a stated number of single and double periods. A square *conflict matrix* whose dimension is the number of lines, is constructed, with a non-zero entry indicating that the corresponding lines are involved in one of the above special requirements. Each line is then given an *availability vector* stating which single and double periods of the week are available to that requirement; this availability vector is also constructed to reflect the balance properties. Starting with a blank timetable, requirements are scheduled according to the following rules:

(i) Select the requirement where the difference between available and required periods is minimal; if this number is zero, assign to the periods available and go to step (iii).

(ii) Where the difference between available and required periods is positive (i.e. there is some choice) a period is chosen to minimize some interference measure which takes account of the reduction in availability to all conflicting lines caused by choosing that period.

(iii) When a period is assigned the availabilities of all conflicting requirements are reduced appropriately. If at any stage the number of periods available is less than the number of periods required, the program has failed.

This program solved a 35 period, 26 class problem in about 90 minutes on a slow machine. Although the program was not really operational, the ideas were important to the development of later systems.

B. The Ontario school scheduling programme (Csima, 1965; Csima and Gotlieb, 1964; Gotlieb, 1963; Lions 1966 a and b; 1967; 1971). The contribution of Gotlieb and Csima was to recognize that the formulation proposed by Appleby, Blake and Newman could be stated more rigorously as a problem in combinatorial mathematics. This led Gotlieb and Csima to search for a set of rules derived from the mathematical structure to replace the heuristic rules used by Appleby, Blake and Newman. The first point to note is that the Ontario approach handled the two kinds of constraints (i)–(vi) and (vii)–(ix) differently. To achieve the 'quality' constraints, they recommended a set of heuristic rules for splitting a weekly problem into the appropriate number of daily problems so as to achieve a desirable spread of subjects over the week. The actual rules and a number of theoretical contributions to the operational system were derived by Lions (1966a) and Griffith (1966) at J. Kates and Associates. To tackle the first set of requirements, Gotlieb realized that essentially simple requirements and preassignments presented the two extremes in terms of difficulties of timetabling; with simple requirements only, timetabling is trivial, while preassignments have absolutely no flexibility in their scheduling. All other requirements could be handled by an appropriate set of preassignments. Thus Gotlieb reduced the timetabling problem to the following combinatorial problem: given a requirements matrix R stating the desired number of meetings on a day between a teacher and a class, and a list of preassignments stating when some of these meetings must take place, does there exist a daily timetable?

The approach taken was similar to the Teddington system; create an *availability array* stating which periods are free for each teacher/class meeting (initially all periods are available). From the list of preassignments (which we can take to be listed in some priority order), sequentially schedule each preassignment, deleting the appropriate availabilities. At each stage the programme checks the current availability array against the remaining requirements matrix to see whether the scheduling of the current preassignment will allow a timetable to be completed. This step represents the heart of the timetabling problem. The check proposed by Gotlieb and Csima was to see whether the availability array created by scheduling the current preassignments satisfies a set of conditions called the *strong Hall conditions* (the timetabling problem is a three-dimensional extension of a classical problem in combinatorial theory – the assignment problem (Kuhn, 1955); the strong Hall conditions are a natural generalization of the conditions for solving the assignment problem). While it was easy to show that the strong Hall conditions were necessary, it was impossible to prove that they were sufficient, although Gotlieb and Csima were prepared to conjecture that this was the case. It has since been shown that the strong Hall conditions are not sufficient (Dempster, 1968; Lions, 1966b); this means that at one stage the algorithm could schedule a preassignment which would not cause the strong Hall conditions to be violated, even though the preassignment was inconsistent with a final timetable. This inconsistency will not be detected until much later in the algorithm, by which time there is nothing to do but go back to the beginning, shuffle the input data, and rerun the program, hoping it does not make the same mistake.

While the theoretical basis of the Ontario programme is thus rather weak, the system nevertheless performs reasonably well in practice. Moreover, the approach taken by Gotlieb and Csima proved to be seminal in the setting up of other systems.

C. The Newcastle system (Barraclough, 1965). The Newcastle system was an extension of the original heuristic approach of Appleby, Blake and Newman, incorporating more sophisticated programming and a second generation computer. In extensive field-trials it showed promise but was not truly operational. Barraclough's formulation of measures of timetabling difficulty and quality, however, have provided yardsticks by which all subsequent systems have been judged.

The second generation systems

These were based on the same approach to the timetabling problem as the first generation systems, but by utilizing more sophisticated heuristics, as well as more powerful machines, they produced the first really operational timetabling systems.

A. The Birmingham system (Clementson 1968). The main feature of this system was an attempt to tackle the underlying formal timetabling problem as a two-dimensional rather than three-dimensional problem. Availabilities were attached to each requirement (in a similar manner to the Teddington and Newcastle systems), but instead of using solely heuristic rules to do the scheduling, a version of the Hall conditions was invoked to check for infeasibilities. Although the programme gave very interesting results on test data, it was found that real school problems were not at all well handled.

B. The Nor-Data system (LGORU, 1972b; Michalson, 1971; Wilson, 1973; 1975). This system began field trials in Norway in 1966 with two schools, and by 1970 was producing 100 schedules a year. It began field trials in UK in 1972, under the aegis of LGORU and has been extensively tested since then. In this respect, it is the first system to achieve a significant degree of success in the UK.

Michalson realizes that the initial requirements are unlikely to be met, and so aims to produce a timetable which is a 'compromise' between conflicting requirements. To resolve these conflicts he distinguishes between absolute requirements, which must be met, and desirable requirements, which he ranks by some priority listing, similar to the Ontario Program. Higher priority requirements take precedence over those of lower priority.

The method of timetable construction includes both combinatorial methods similar to Clementson's version of the Gotlieb-Csima work, and heuristic rules to allow for the fact that the strong Hall conditions are not sufficient. These heuristics are what Michalson describes as 'local conditions', being a mixture of local versions of the strong Hall conditions, probabilistic evaluations and other heuristic rules. These are used together with restrictions on the 'quality' of the timetable (e.g. distribution quality) so that the programme aims to maximize the probability of completing a timetable while meeting as many as possible of the qualitative requirements.

The complexity and size of some modern comprehensive schools poses a severe test for any timetabling system, and the ILEA report (Wilson, 1975) states that the two year trials with Nor-Data suggest that the larger and more complex the school, the less chance there is of a successful timetable being produced, and concludes: 'The 1974 Nor-Data trials suggest that ILEA should continue testing the system . . . in the hope that, in time, the tests can be extended to the schools who are most in need of computer-aided timetabling'.

On some of the ILEA schools it was found necessary to tackle the timetabling in stages, doing the upper school first, which is probably the most complex and least flexible part of the timetable, and then modifying the requirements of the lower school to fit in with this timetable. In this respect the Nor-Data system is moving towards a more interactive approach in which the timetabling and planning stages are intermixed. It is this kind of approach which we shall discuss in more detail later.

C. The New Zealand system (LGORU, 1972a; NCCL, 1973) While this system has not been as extensively tested as the Nor-Data system, it has achieved a considerable degree of success in the UK and it is now being offered as a commercial system. Not much is known about the details of its operation, but it appears to use the usual mix of combinatorial and heuristic rules – selecting periods for requirements to satisfy constraints on distribution through the week, staff availability, etc; if any choice is left, a period is selected which will least likely preclude later assignment of periods. As in the case of the Nor-Data system, its ability to produce acceptable timetables, at least for some schools, is its best recommendation.

Two other systems are known to us, although we have no published information concerning them.

D. The Price-Waterhouse system (1971) We believe this to be similar to the Nor-Data and New Zealand systems in using a mix of combinatorices and heuristics. It has had limited field trials in the U.K., with some favourable reports, particularly for some rather large and complex schools, but we understand that it is no longer being offered to local authorities.

E. The Scicon system – B. Millane (1972) This was a modification of a resource allocation routine of a network analysis computer program which had been developed initially by M. Beale for industrial applications. Work ceased when the best percentage fill obtained (said to be 85 per cent) proved to be far too low for acceptable use in schools.

The third generation approaches

The systems studied so far have all been based on the conventional view of the timetabling process – the headmaster presents the timetabler with a complete list of requirements and the timetabler tries to construct a timetable incorporating as many of the requirements as are compatible. As we pointed out earlier, in practice a headmaster would find it very difficult to provide a complete specification of the requirements he wishes for a timetable, for three reasons. First, many of the requirements on a timetable are rather subtle ones which reflect the particular circumstances of the school – the preferences of a particular member of staff, the geographical layout of the school, the particular arrangements for assembly, lunch, etc. The headmaster will not specify these, simply because he expects his timetabler to be aware of these factors and build them into the timetable. Secondly the headmaster does not want to make excessively detailed choices before the timetable is drawn up, because he knows that this may prevent a timetable from being constructed. He is prepared to leave many decisions to the timetabler, so that the timetabler can choose whatever is most convenient for producing a timetable. An obvious example is that headmasters may not assign specific teachers to teach specific classes, especially at the lower end of the school, and the timetabler will select whichever teachers have most free periods left when he comes to timetable these classes. However, the choice will not be made solely on the grounds of timetabling convenience, but will also be consistent with the school's educational objectives. Thirdly, the initial requirements will change as staff availability or pupil choice change.

Thus it is unlikely that a timetable produced to meet the headmaster's initial set of requirements, would be satisfactory, either because the initial requirements are mutually incompatible, or because the timetable fails to meet some unstated requirements. It is important, therefore, that a timetabling system should be flexible enough to allow the headmaster to modify the requirements and see what effect this has on the timetable.

Of course, any computer timetabling system offers this flexibility in a spurious way, since it is always possible to change the requirements and rerun the programme to produce a new timetable. But this is unsatisfactory; firstly because even with the power of modern computers most systems will require at least about ten minutes to compute a school's timetable, and several runs of that length become very expensive. More important, it can mean that a few changes in the requirements can lead to a markedly different timetable being produced.

Thus in order to provide the system with the required flexibility, it is necessary to design the system to meet this possibility. The systems described in this section have been so designed, and thus are aimed less at the stage of detailed construction and more at the overall problem of curriculum planning. However, they have still to prove to be operationally viable.

A. The Strathclyde system (Laurie, 1969; Lawrie and Veitch, 1973). After a careful study of how schools actually construct their timetables, Lawrie noted that at the end of the first stage of timetabling headmasters had not prepared a complete list of requirements, only an outline plan. Lawrie's procedure is aimed at helping the headmaster at this stage of the planning process rather than at the stage of detailed timetable construction. The problem is not formulated using classes and teachers as the basic units, but rather year groups and departments. While this suppresses a considerable amount of detail, Lawrie argued that this formulation of the problem was of considerable importance. This was especially true at the time Lawrie began work, for schools were involved in quite drastic organizational changes – the move to comprehensive schools and raising of the school leaving age. The headmaster was concerned more with checking the overall feasibility of alternative organizational patterns, for example different arrangements of option grouping, than with detailed scheduling of individual classes and teachers. Thus the emphasis of Lawrie's system was towards curriculum planning, the first stage of timetabling, rather than to the second stage, detailed timetable construction. This had the advantage of not requiring that the headmaster prepare a more fully specified statement of requirements than that to which he was accustomed.

There are two parts to Lawrie's program. In the first, the requirements of different year groups are fitted together so as to ensure that each department is not overloaded. Lawrie calls this an 'outline timetable', although it is not really a timetable, in the sense that the groups of requirements have not been assigned to periods. This is handled in the second stage of the programme where the columns of the outline timetable are permuted so as to achieve a reasonable distribution of requirements through the week.

Initial trials suggested that the first part of the programme, based on familiar integer linear programming routine, was successful, but in later, more extensive trials, it was found that outline timetables could not always be produced, due to infeasibilities in the specification of requirements. However, the programme was not able to pinpoint the source of the incompatability. Moreover, the familiar problems of integer programming routines often resulted in unacceptably long run times.

Even where the first part of the programme worked to produce acceptable outline timetables, the second part of the programme, which allocated requirements to specific hours of the week, never operated very successfully, so that in most cases schools were left simply with outline timetables. The result was that schools were still left with a considerable amount of work to actually produce a timetable, for they would have to do the distribution of periods by hand, and then assign specific teachers to specific classes, and in some cases the structure of the model Lawrie used might prevent this being done in a satisfactory manner; specifically it could result in a class being taught the same subject by two different teachers during the week. This is hard to prevent in a programme which suppresses individual teacher identities.

Thus the Strathclyde system has not proved to be operationally viable, but it is nonetheless important for providing a new approach to the timetabling problem.

B. The Oxford system (Dempster, 1965; 1968; 1971; Early, 1968) As a co-worker on the Ontario School Scheduling Program, discussed earlier, Dempster based his system on the Gotlieb-Csima framework – the weekly problem being split into daily problems, which are then treated as a requirements matrix with an appropriate list of preassignments. However, Dempster noted that the strong Hall conditions used by Gotlieb and Csima to check on infeasibility were not sufficient, and devised a different approach to the problem. Instead of starting with a blank timetable and inserting each requirement in turn, checking for infeasibility at each step, he started with a timetable which satisfies all the simple requirements, and modified it to include the preassignments. The initial timetable could be either last year's timetable (or as much of it as is consistent with this year's timetable), or one that is generated automatically from the requirements matrix, since this is easily done. The process of altering a complete timetable so that it incorporates the current preassignment is called an interchange, and the important contribution made by Dempster was that he derived necessary and sufficient conditions for such an interchange to be feasible. Thus, for a daily problem, defined in the specialized way used by Gotlieb and Csima, if a timetable exists Dempster's algorithm must find it, while if the preassignments are incompatible, the algorithm can pinpoint the source of incompatibility.

From the point of view of constructing a timetable from an initial list of requirements, Dempster's method suffers from two problems. Firstly it requires all requirements to be translated into a list of preassignments, which it is not possible to do rigorously, in the sense that the choice of preassignments for one kind of requirement may preclude a sensible choice for a later requirement. Secondly, in practice, the introduction of extensive setting in schools, particularly in the upper school, leads to very complex sets of preassignments, and the time taken to interchange from some initial timetable can become unacceptable. In order to construct a timetable from a given set of requirements, therefore, the Oxford system has used a simpler form of interchange, based on the ideas of Matula (Matula *et al*, 1972). This is now an operational system, and some results from the first trials are available (OSA, 1975).

However, the notion of modifying an existing complete timetable to incorporate new requirements, where some part of the existing timetable is fixed, lends itself very naturally to the problem of providing a flexible timetabling system, for the interchange algorithm can incorporate new requirements into a timetable (or simply rearrange the existing requirements) with minimal interference to the existing timetable. The Oxford system envisages a fully interactive system where the headmaster specifies the basic requirements for the timetable and an initial timetable is generated; either because of incompatabilities in the original data, or changing circumstances, this timetable is unlikely to be completely satisfactory, and the headmaster can then 'play' with the timetable, taking out some requirements and inserting others.

However, this interactive system is not yet operational. It remains an attractive and promising idea.

Conclusions

We have presented a brief, non-rigorous description of the way computer timetabling systems have evolved in the UK. We believe that timetabling is an important and difficult problem, and this is likely to become increasingly so with larger schools and resource shortages. In addition we have tried to demonstrate that the differences between the systems described are as much due to differing perceptions of the problem as to any technical or programming variations.

We have tried, deliberately, to avoid any evaluation of the various systems, beyond pointing out those which have proved operational and those which have not. In this respect, only the two second generation systems – Nor-Data and SPL – currently offer well-tested timetabling services, although the Oxford system is now beginning trials. Readers seeking evaluation are referred to the appropriate reports (LGORU, 1972a and b; Wilson, 1973), although more up-to-date reports are becoming available. In practice, the qualitative judgement of the individual schools themselves must be the ultimate arbiters of the merits or otherwise of any system.

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